

EEE109 Lab 2 – BJT and Amplifiers

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Abstract

This experiment used the NI ELVIS III platform and a 2N3904 bipolar junction transistor to study the basic characteristics of a BJT and its use in switching and amplifier circuits. The work included observing transistor output characteristics, studying cutoff, saturation, and active-region behavior, measuring the DC bias point, voltage gain, and frequency response of a common-emitter amplifier, checking the voltage-following and current-gain behavior of an emitter follower, and completing a combined design task. The experiment involved breadboard construction, DC measurements, oscilloscope observation, and comparison between theoretical and measured values. The results were generally consistent with transistor theory. The common-emitter circuit showed clear voltage gain with phase inversion, the emitter follower had a voltage gain close to 1 with good current driving ability, and the final design task showed clear switching between different output states.

1 Introduction

1.1 Overall Objective

This experiment was the second lab in EEE109 and focused on BJTs and amplifiers. Using the NI ELVIS III platform and a 2N3904 transistor, the lab was divided into five parts: observing transistor output characteristics, studying switching behavior, analyzing the gain and frequency response of a common-emitter amplifier, checking the behavior of an emitter follower, and completing a comprehensive design task. The main aim was to strengthen understanding of transistor biasing and amplification and to connect theory with practical measurements.

1.2 Theoretical Background

An NPN bipolar junction transistor has three terminals: base, emitter, and collector. In the active region, the base-emitter junction is forward biased and the collector-base junction is reverse biased. In this condition, the collector current is mainly controlled by the base current and can be described by the current gain β . A transistor usually works in cutoff, satu-

ration, or active region, which correspond to off, fully on, and linear amplification.

A common-emitter amplifier needs a proper bias network to set a stable operating point, while coupling and bypass capacitors affect the frequency response. Its output is usually inverted and has noticeable voltage gain. An emitter follower gives an output voltage close to the input voltage, only lower by about one V_{BE} , so it has high input impedance, low output impedance, and a voltage gain close to 1. This makes it useful as a buffer stage.

1.3 Expected Behaviour of the Circuits

In Section 1, the collector current of the BJT was expected to become more stable in the active region as the collector voltage increased, and different base currents should produce different characteristic curves. In Section 2, the transistor was expected to show clear cutoff and saturation behavior as a switch. In Section 3, the common-emitter amplifier should give an inverted output and a fairly stable gain in the mid-frequency range. In Section 4, the emitter follower should give an output that follows the input with a voltage gain close to 1. In Section 5, the final design circuit should show clearly different output responses in different operating states.

1.4 The Organization of the Report

This report begins with an Abstract that summarizes the aim, method, results, and conclusion of the experiment. The Introduction presents the background, theory, and expected circuit behavior. Experimental Setup and Procedures describes the setup and procedure, Results and Discussion presents and analyzes the results, and the Conclusion summarizes the main findings.

2 Experimental Setup and Procedures

2.1 Instruments and Components

All circuits in this experiment were built on the NI ELVIS III proto-board. This platform was the main piece of equipment because it includes a DC power supply, digital multimeter, function generator, oscilloscope, and BJT measurement tools, which made it convenient to build and test circuits in one place. The main active device used in Sections 1 to 4 was the 2N3904 NPN transistor. Other components included fixed resistors, coupling and bypass capacitors, variable resistors, and diodes used in the final design task.

Before each part of the experiment, the circuit was connected according to the given schematic and the component values were checked carefully to reduce wiring mistakes. The transistor pin order was also checked each time, because reversing the collector, base, and emitter would make the measurement results meaningless.

2.2 Section 1: Bipolar Junction Transistor Characteristics

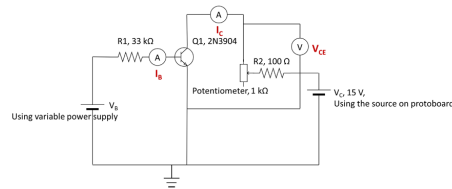


Figure 1: BJT characteristic measurement circuit

2.2.1 1.1 BJT Behavior

In Section 1.1, the IV Analyzer in NI ELVIS III was used first to observe the basic output characteristics of the 2N3904 transistor. The transistor type was set to NPN, the collector resistance setting was low, the collector voltage sweep was set from 0 to 3 V with a step of 0.1 V, and the current

limit was 30 mA. The base current sweep started at $15\text{ }\mu\text{A}$ with a step of $11\text{ }\mu\text{A}$ for 5 curves. After the settings were completed, the IV Analyzer was run and the output characteristic graph was saved for later analysis.

2.2.2 1.2 BJT Measurements in the Active Region

In Section 1.2, the active-region test circuit from the lab script was built manually on the breadboard and measured using a digital multimeter and a variable DC power supply. The base voltage was set to 2.38 V, 4.08 V, and 5.76 V, which corresponded to about $I_B = 50\text{ }\mu\text{A}$, $100\text{ }\mu\text{A}$, and $150\text{ }\mu\text{A}$. For each base bias condition, the potentiometer was adjusted step by step so that V_{CE} took values from 6 to 12 V, and the corresponding collector current I_C was recorded. These data were then used to plot the relationship between I_C and V_{CE} .

During the actual setup, the breadboard power ground, measurement ground, and signal ground had to be connected together. Otherwise, the readings were not stable. After the wiring was checked, the data for each case were measured and plotted.

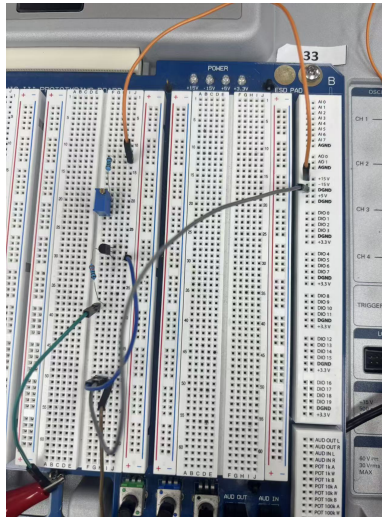


Figure 2: Practical circuit setup for Section 1 transistor characteristic measurement

2.3 Section 2: BJT as a Switch

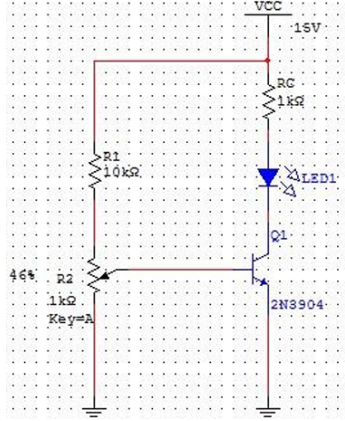


Figure 3: Transistor switch circuit schematic

This section is based on calculation using the given circuit diagram and parameters, so no extra circuit build or measurement was required. The values to be found are V_{CE} at cutoff, V_{CE} at saturation, and the saturation values of V_{RC} and I_C . The calculation conditions are $V_{CC} = 15\text{ V}$, LED drop 1.8 V , and $V_{CE(\text{sat})} = 0.1\text{ V}$.

2.4 Section 3: Common Emitter Amplifier

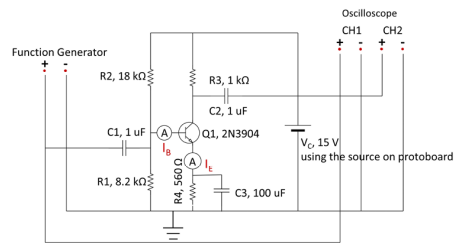


Figure 4: Common-emitter amplifier circuit schematic

2.4.1 Transistor Amplifier

In Section 3, a common-emitter amplifier was built. The bias resistors, collector resistor, emitter resistor, input coupling capacitor, and emitter bypass capacitor were connected according to the circuit diagram. After the circuit was built, an AC input signal was not applied immediately. Instead, the DC operating point was checked first. The base, collector, and emitter voltages were measured with the digital multimeter to confirm that the transistor was working in the active region. The collector voltage was checked carefully to see whether it was roughly in the middle of the supply range, since this allows a larger output swing and helps avoid early clipping.

2.4.2 Frequency Response of the Amplifier

After confirming the bias condition, a small sinusoidal signal was applied to the input through the coupling capacitor. The input and output waveforms were observed using the oscilloscope. The voltage gain was estimated by comparing the amplitudes, and the phase relation was checked because a common-emitter amplifier should normally invert the output. For the frequency response, the input frequency was changed gradually while keeping the signal small enough to avoid distortion, and the output amplitude at low, mid, and high frequencies was observed.

During testing, too many jumper wires were found to cause poor contact and reduce measurement reliability. Therefore, the wiring should be kept simple, and long unnecessary connections should be avoided. It was also important to prevent resistor and capacitor leads from touching each other or causing short circuits.

2.5 Section 4: Emitter Follower Amplifier

In Section 4, the basic operation of the emitter follower was first considered from the circuit model, and then the actual circuit was built on the protoboard using the transistor, bias resistors, and the required input and output connections. Compared with the common-emitter amplifier in the previous section, the main point here was not to obtain large voltage

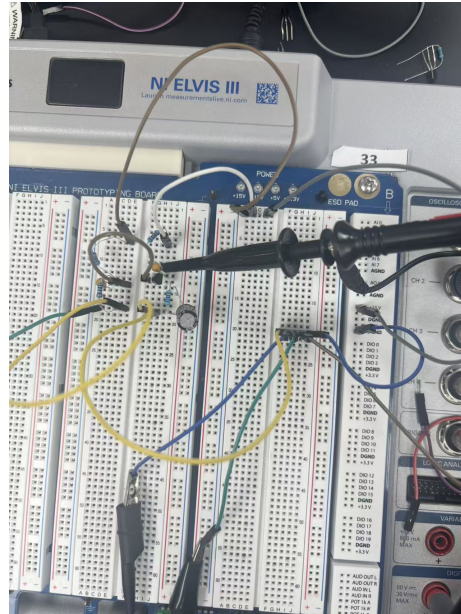


Figure 5: Practical setup of the Section 3 common-emitter amplifier

gain, but to observe how the circuit follows the input voltage and works as a buffer stage.

After the circuit was connected, the DC voltages at the base and emitter were measured first to check whether the emitter voltage followed the base voltage with about one base-emitter junction drop difference. After confirming that the DC operating condition was normal, a sinusoidal signal was applied to the input and the output was taken from the emitter. The input and output waveforms were then compared using the oscilloscope. The main things checked were that the output was not inverted, the voltage gain was close to 1, and the load-driving ability was stronger than that of the common-emitter amplifier.

2.6 Section 5: Comprehensive Design Task

In Section 5, the comprehensive design task combined the ideas of amplification, coupling, biasing, and driving from the earlier parts into one circuit. The focus here was no longer on the gain of a single stage, but on

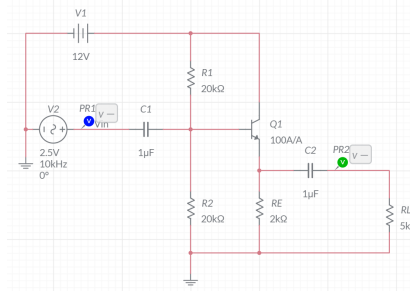


Figure 6: Section 4 emitter follower circuit schematic

checking whether the whole system could work properly and produce clearly different outputs under different conditions.

2.6.1 Designed Schematic

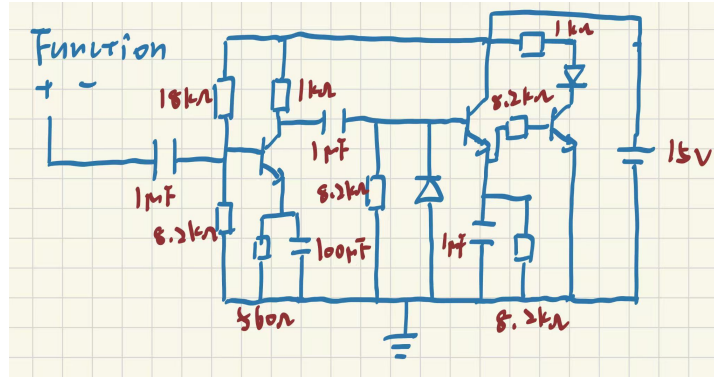


Figure 7: Schematic of the Section 5 comprehensive design task

The circuit primarily consists of three parts: the first stage is a common-emitter amplifier, comprised of an input coupling capacitor, bias divider resistors, collector resistor, emitter resistor, and bypass capacitor, which serves to amplify the weak input signal; the second stage is a coupling and control network that sets the turn-on condition through resistors and a diode, and couples the AC signal from the previous stage to the following stage; the third stage is the output drive and load section, where, upon meeting the trigger condition, the subsequent transistor stage and the load

branch generate a clear output signal.

3 Results and Discussion

3.1 Section 1: Bipolar Junction Transistor Characteristics

3.1.1 1.1 BJT Behavior

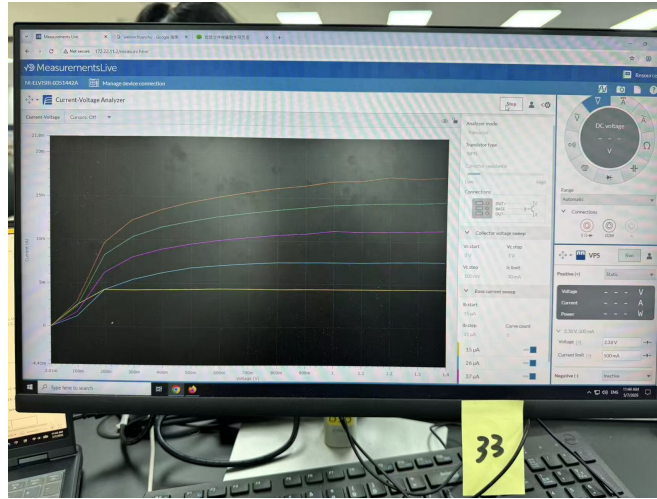


Figure 8: Section 1.1 BJT output characteristic measurement

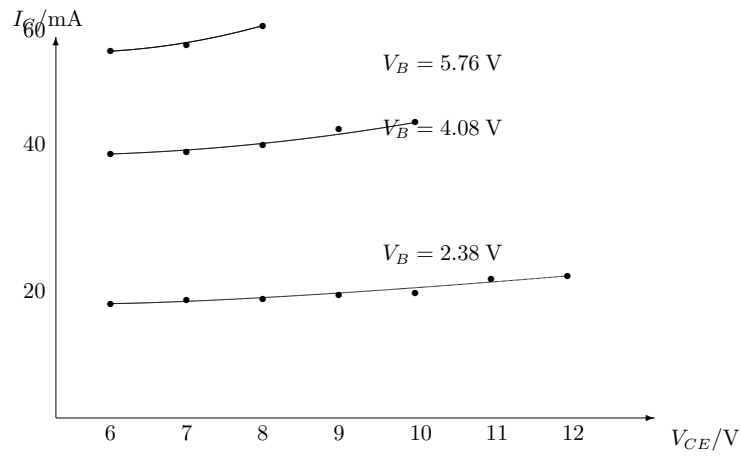
The output characteristic graph generated by the IV Analyzer shows that, as the base current increases, the I_C - V_{CE} curves move upward. This means that a larger base drive produces a larger collector current. At low V_{CE} , the curves rise quickly. As V_{CE} increases further, the curves become flatter, which shows that the transistor is in the active region and the collector current is mainly controlled by the base current rather than by V_{CE} . This result agrees with the expected BJT output characteristics.

3.1.2 1.2 BJT Measurements in the Active Region

In the manual measurement part, V_B was set to 2.38 V, 4.08 V, and 5.76 V, and the corresponding I_C values were recorded for $V_{CE} = 6$ to 12 V

Table 1: Measured data in the active region

V_{CE} (V)	I_C at $V_B = 2.38$ V (mA)	I_C at $V_B = 4.08$ V (mA)	I_C at $V_B = 5.76$ V (mA)
6	18.6	39.5	53.8
7	19.1	39.8	54.6
8	19.3	40.7	57.3
9	19.8	42.9	N/A
10	20.0	43.8	N/A
11	21.9	N/A	N/A
12	22.3	N/A	N/A

Figure 9: I_C - V_{CE} graph plotted from Table 1

in each case. From Table 1 and the plotted graph, it can be seen that for the same base bias, I_C does not change much with V_{CE} . This shows that the transistor was basically working in the active region at these measurement points. When the base bias increased from 2.38 V to 5.76 V, the whole curve moved upward, which means that the collector current increased with the base current.

Looking at the overall trend, β was not exactly constant at every point, but it did not change too much within the same operating region. The difference may come from transistor parameter variation, manual adjustment error, breadboard contact quality, and instrument resolution. Overall, the automatic test and the manual test in Section 1 support each other and show the BJT behavior in the active region clearly.

3.2 Section 2: BJT as a Switch

Table 2: Calculated results for the transistor switch

Data	Calculated Values
V_{CE} (cut-off)	15.0 V
V_{CE} (saturation)	0.1 V
V_{RC} (saturation)	13.1 V
I_C (saturation)	13.1 mA

From the table, it can be seen that at cutoff the transistor is basically off, so V_{CE} is close to the supply voltage. In saturation, V_{CE} drops to about 0.1 V and the collector current is about 13.1 mA. This shows that the circuit has clear switching behavior.

3.3 Section 3: Common Emitter Amplifier

3.3.1 Transistor Amplifier

Tables 3 and 4 show that the calculated values and measured results of the common-emitter amplifier are fairly close. Because the base current was very small, it was not convenient to measure directly. So $\beta \approx 97.49$

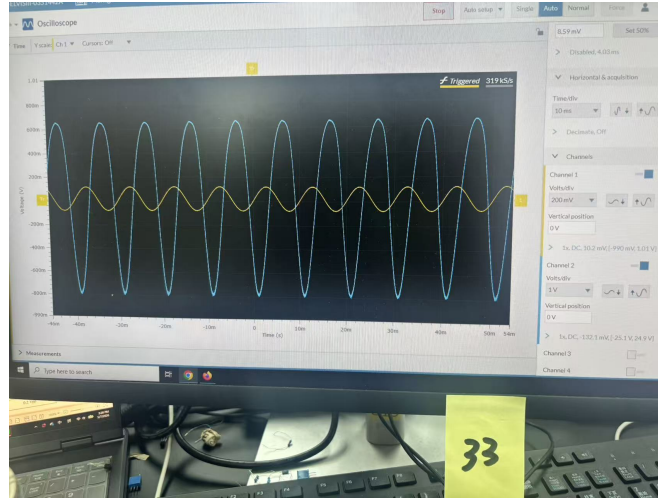


Figure 10: Input and output waveforms of the common-emitter amplifier

Table 3: DC bias results

Data	Calculated Value	Measured Value
I_B	$64.2 \mu\text{A}$	—
I_E	6.49 mA	6.323 mA

Table 4: Comparison of transistor parameters

Data	Given Value	Measured/Calculated Value
β	100	97.49
V_{BE}	0.7 V	0.6713 V

was estimated from the calculated value of I_B and the measured value of I_E . This is close to the nominal value of 100, which suggests that the bias design was reasonable.

3.3.2 Frequency Response of the Amplifier

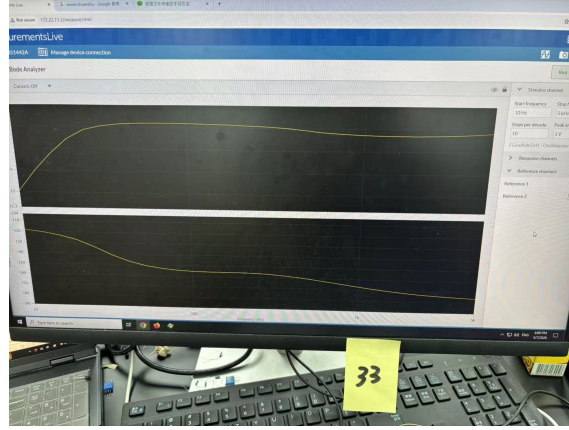


Figure 11: Measured frequency response of the common-emitter amplifier

From the oscilloscope data, the input peak-to-peak voltage was about 201.9 mV and the output peak-to-peak voltage was about 7.135 V, so the voltage gain was

$$A_v = \frac{7.135}{0.2019} \approx 35.34.$$

which gives, in decibels,

$$20 \log_{10}(35.34) \approx 30.96 \text{ dB}.$$

From the Bode Analyzer result, the gain at about 100 Hz was around 16.15 dB. Based on the small-signal model, the theoretical gain near 100 Hz was about 34.2 dB, which is closer to the oscilloscope result but quite different from the Bode Analyzer reading. This difference was most likely caused by the Bode test amplitude, the selected measurement points, or the instrument settings. Overall, the common-emitter amplifier clearly produced voltage amplification and an inverted output.

3.4 Section 4: Emitter Follower Amplifier

3.4.1 Waveform Observation

After the AC input signal was applied, the input and output waveforms were almost in phase, and the output amplitude was slightly smaller than the input. This shows the typical behavior of an emitter follower. In other words, the voltage gain of the circuit is close to 1, but it can follow the input signal well, so it is suitable for use as a buffer stage.

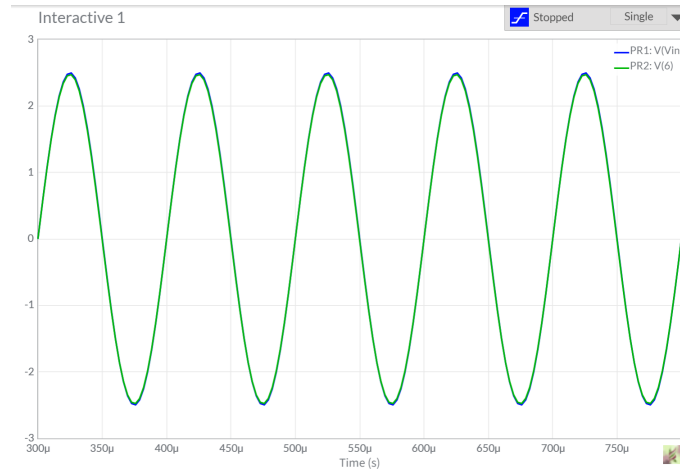


Figure 12: Input and output waveforms of the emitter follower

3.4.2 Current Gain

Table 5: Current gain for different R_E values

Resistance of R_E	Probe 1 – Input Current	Probe 2 – Output Current	Gain
1 k Ω	58.689 μ A	987.82 μ A	16.83
2 k Ω	34.224 μ A	986.60 μ A	28.83
3 k Ω	26.035 μ A	985.86 μ A	37.87
5 k Ω	19.385 μ A	977.07 μ A	50.40

Table 5 shows that, under different R_E conditions, the output current stayed around 0.98 mA while the input current was only a few tens of mi-

croamps. This means that the circuit had a clear current gain. As R_E increased, the input current decreased and the current gain became larger. This agrees with the fact that the emitter follower works well as a current buffer stage.

3.5 Section 5: Comprehensive Design Task

Section 5 combines the amplification, coupling, and driving circuits from the previous sections. Instead of focusing on one stage only, this part looks at the output difference between the untriggered and triggered states to judge whether the whole circuit can perform the expected function in a stable way.

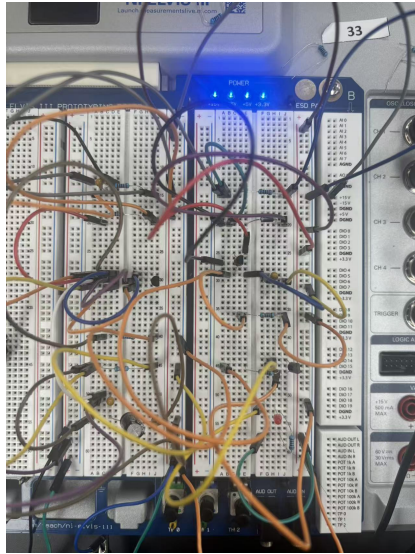
3.5.1 Circuit Behaviour

The idea of this circuit is that the front stage first amplifies the input signal, and then the signal is passed to the later stage through coupling so that the conduction state of the later stage can be controlled. In other words, the front stage provides enough signal amplitude, and the later stage decides whether a clear output will appear. So, in the untriggered state, the output should stay low, while in the triggered state, the output should change clearly.

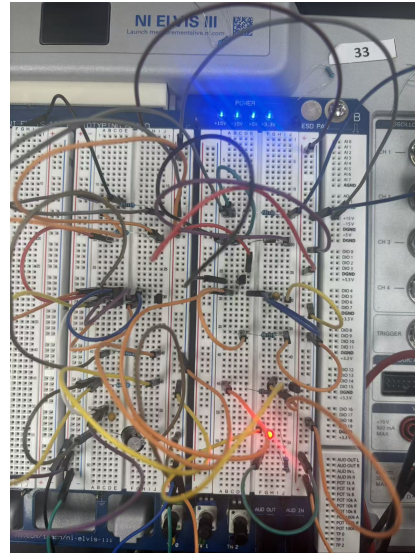
3.5.2 Measured Output States

To compare the outputs in the two states, the oscilloscope and digital multimeter were used to record the results. In the untriggered state, the output shown in Fig. 14 had almost no clear AC signal on the oscilloscope, and the multimeter reading was about 2.422 mV. This means that under the untriggered condition, the input signal was not effectively passed to the final output stage and the later stage was basically not driven.

When the circuit entered the triggered state, the output shown in Fig. 15 had a clear sinusoidal waveform on the oscilloscope, and the multimeter reading was about 1.8853 V. Compared with the untriggered state, the output increased a lot, which means that the front-stage signal was passed



Untriggered state



Triggered state

Figure 13: Practical circuit photos for Section 5

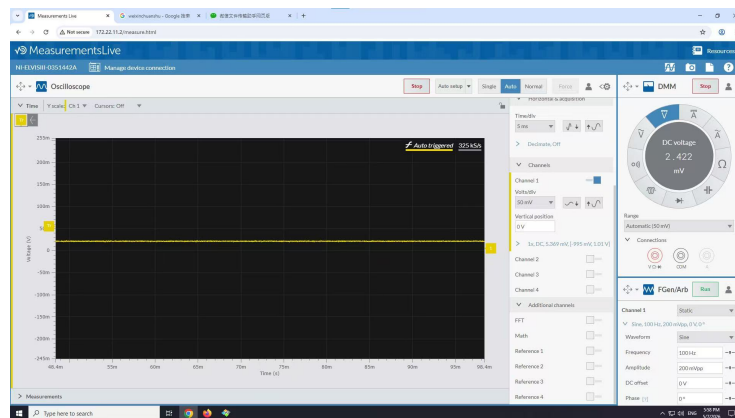


Figure 14: Output of the comprehensive design task in the untriggered state

effectively to the later stage and produced a clear response there.

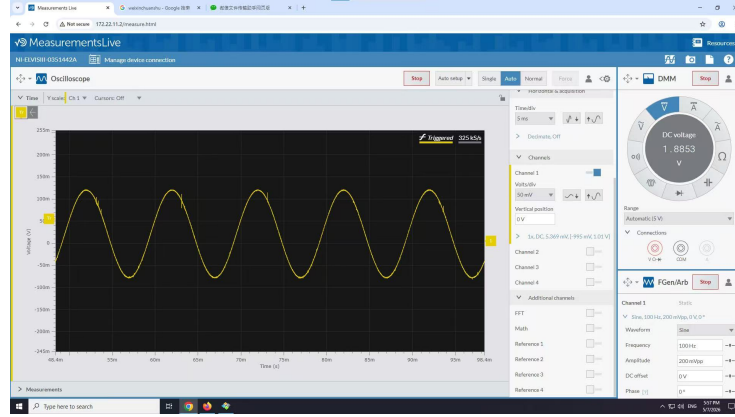


Figure 15: Output of the comprehensive design task in the triggered state

From Fig. 14 and Fig. 15, it is clear that the circuit gave very different outputs in the two states. In the untriggered state, the output was close to zero, while in the triggered state, a clear AC signal and a higher output voltage appeared. This shows that the front-stage amplifier, coupling network, and later driving stage were able to work together, so the overall circuit reached the expected goal.

4 Conclusion

This experiment studied the basic characteristics of a bipolar junction transistor, its switching use, and its amplifier applications. Sections 1 and 2 showed the output characteristics of the transistor and its cutoff and saturation behavior as a switch. The results were generally consistent with the expected theory.

In the amplifier part, the common-emitter circuit in Section 3 showed clear voltage gain and phase inversion, which means the circuit could amplify effectively. The estimated value of $\beta \approx 97.49$ was also close to the nominal value. The emitter follower in Section 4 showed a voltage gain close to 1, an in-phase input and output, and a noticeable current gain, which shows that it works well as a buffer stage.

Section 5 further showed that the front-stage amplifier, coupling network, and later driving stage could work together in one system and produce clearly different outputs in different states. Overall, the experiment agreed well with the theoretical BJT model. Although there were still effects from component tolerances and wiring conditions, the main results were clear and easy to observe.